

Enrique Martínez Martel

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SUMMARY

Data Scientist with a background spanning computer vision, NLP, and applied ML across academic research and industry. Experience building end-to-end ML systems, from object detection models for real-world environments to data pipelines and LLM-based classification workflows.

EXPERIENCE

Banco Santander Jan. 2025 – Present
Data Scientist - IRB Risk Madrid, Spain

- Part of the IRB Risk team; involved in the generation of regulatory credit risk parameters PD (Probability of Default) and LGD (Loss Given Default) under the Basel framework.
- Developed and maintained ETL pipelines and data generation workflows in Databricks (Python/PySpark); provided ad-hoc support to the LGD team including outlier investigation and preparation of responses to ECB and Banco de España inspections.

Banco Santander Jun. 2024 – Dec. 2024
Data Science Talent Program - Models & Data Madrid, Spain

- Designed and benchmarked an LLM-based email classification pipeline using GPT-4/GPT-4o via Azure OpenAI; evaluated multiple prompting strategies on real operational data.
- Supported the Mortgages business team through customer segmentation, EDA, and automated reporting using Python.

Omron Industrial Automation Europe Jan. 2022 – Aug. 2023
Junior Software Engineer Barcelona, Spain

- Contributed to software solutions optimising industrial automation processes using C#/.NET, Visual Studio, and TFS.

PROJECTS

MSc Thesis: Comparative Analysis of Computer Vision Techniques UPM - 2025

- Comparative study of object detection architectures (YOLOv10, Faster R-CNN) on a custom dataset of urban accessibility features; identified YOLOv10L as optimal for real-time deployment. Thesis grade: 10/10.

BSc Thesis: Autonomous Driving with CARLA UPC - 2021

- Developed a perception and decision-making pipeline for autonomous driving using the CARLA simulator; covered sensor data processing, object detection, and real-time control under dynamic environments.

EDUCATION

Universidad Politécnica de Madrid (UPM) Sep. 2023 – Jun. 2025
MSc in Data Science [Thesis Grade: 10/10] Madrid, Spain

Escuela de Organización Industrial (EOI) Sep. 2024 – Jun. 2025
MSc in Big Data & Digital Business Madrid, Spain

Universitat Politècnica de Catalunya (UPC) Sep. 2017 – Jun. 2021
BSc in Computer Science Barcelona, Spain

SKILLS

AI/ML : LLMs, Prompt Engineering, XGBoost, LightGBM, PyTorch, Transformers, YOLO, Scikit-learn, Streamlit

Languages : Python (PySpark, Pandas, NumPy), SQL, C#/.NET

Tools & Platforms : Apache Spark, Databricks, Azure OpenAI, Git/GitHub, Power BI, Linux, Azure DevOps

Spoken Languages : Spanish (native), English (full professional proficiency), French (basic)